Module 4

Continuous-Time Dynamic Programming and Reinforcement Learning

In this document we present several MATLAB and Simulink engineering design projects relevant to dynamic programming and reinforcement learning in the continuous-time domain according to the material presented in Modules 1-4.

MATLAB Reinforcement Learning Projects

Project 4.1. RL Algorithm for the Electric Islanded Microgrid System

The project topic is the reinforcement learning mechanism (RL reinforcer) of the linear-quadratic (LQ) optimal control problem in continuous time. The project design requires the use of MATLAB/Simulink including live scripts and live functions. The students are expected to demonstrate the RL solution (see block diagram in Figure 1.1 in Module 1) to the LQ optimal control problem in the continuous-time domain using the algorithm presented in Section 4.4.

Project Data: The project considers an islanded electric microgrid system whose continuous-time space matrices are presented in Homework Problem 2.9 of Module 2. Assume that the penalty matrices in the corresponding LQ optimal control problem are identities, that is, given by $Q = I_4, R = I_2$.

Project Tasks: Check that the assumptions for the existence of the LQ optimal control are satisfied. Make a table that shows how the approximate performance values converge per iteration to the optimal values. In general, you need to run only five to six iterations, in rare case nine or ten iterations since the Lyapunov iterations are in fact the Newton method iterations for solving the algebraic Riccati equation, which will be chosen in the remaining part of this project assignment. It is known that the Newton method has the quadratic rate of convergence. Plot on one figure all approximates feedback controls obtained for $k = 1,2,3,4,5$, and on another figure plot the approximate trajectories for a selected (the one that performs worse in your opinion) state variable for $k = 1,2,3,4,5$.

Project Report Format: The project report is supposed to be typed and it must contain all standard technical paper main parts: introduction (about the general methodology used), problem formulation (what you plan to do and what you are supposed to achieve) main results obtained (obtained simulation results, tables, and figures), discussion/comments (on obtained results and

some observations), conclusions, appendix (with MATLAB code placed in it), references. The report should contain a brief description of the policy iteration algorithm with analytical presentation of all important formulas, equations, required conditions (assumptions).

Project 4.2. RL Algorithm for the Wind Turbine System

A continuous-time linear wind turbine mathematical model state space matrices are presented in Homework Problem 2.10 in Module 2. Apply the Vaisbod-Milstein-Kleinman reinforcement learning algorithm following the project formulation and data presented in Homework Problem 2.10 in Module 2 and Design Project 3.1 of Module 3. Write a live script code with corresponding live functions.

Project 4.3. RL Algorithm for the Satellite Optimal Control Problem

Consider the satellite optimal control problem whose continuous-time state space matrices are given in Problem 2.12 in Module 2. Repeat all steps outlined of Design Project 3.1 from Module 3 for reinforcement learning optimal control of the satellite mathematical model. Write a live script code with corresponding live functions.

Project 4.4. RL Algorithm for the F-15 Aircraft Optimal Control

Consider the optimal control problem of F-15 aircraft whose continuous-time state space matrices are given in Homework Problem 2.11 in Module 2. Repeat all steps outlined of Design Project 3.1 from Module 3 for reinforcement learning optimal control of this aircraft. Implement the reinforcement learning mechanism according to Module 1 Figure 1.1 using Simulink.

Project 4.5. RL Algorithm for a Drone Yaw Angle Optimal Control

Consider a drone yaw angle control continuous-time transfer function considered in Exercise 2.1 in Module 2. Obtain its state space form. Repeat all steps outlined of Design Project 3.1 from Module 3 for reinforcement learning optimal control drone yaw angle control. Implement the reinforcement learning mechanism according to Module 1 Figure 1.1 using Simulink.